ORDER
7110.11B

Effective Date:
March 1, 2026

SUBJ: HLN ATCT/APPROACH Standard Operating Procedures

This document establishes the Helena Airport Traffic Control Tower (HLN ATCT) and the Helena Airport Non-Radar Approach Control (HLN APP) Standard Operating Procedures within the Salt Lake City ARTCC on VATSIM (vZLC). This SOP establishes standardized air traffic control, flight operations, and airfield management procedures at Helena (HLN). It ensures safe, efficient, and coordinated air traffic operations by detailing responsibilities, communication protocols, and operational guidelines. The provisions and procedures described herein are supplemental to vZLC Facility Policies and FAA Order JO 7110.65.

The information contained herein will be used only for flight simulation purposes on the VATSIM network. It is not intended, nor should it be used for real-world navigation. The Virtual Salt Lake City ARTCC is not affiliated with the FAA, Salt Lake City ARTCC, or any governing aviation body.

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Record of Changes

Version	Description	Effective Date	Issued By
7110.11A	Initial Release In New Format	Feb 10, 2026	CN
7110.11B	Fixing Typos and adding Appendices 1-4	March 1, 2026	CN

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Chapter 1. General Control

Section 1. Introduction

1-1-1. Purpose
This document establishes the Helena Air Traffic Control Tower And Approach Control Facility (HLN ATCT/APPROACH) Standard Operating Procedures within the Salt Lake City ARTCC on VATSIM (vZLC). This SOP establishes standardized air traffic control, flight operations, and airfield management procedures at Helena (HLN). It ensures safe, efficient, and coordinated air traffic operations by detailing responsibilities, communication protocols, and operational guidelines.

1-1-2. Audience
All vZLC controllers and visitors contained within the vZLC and VATUSA roster.

1-1-3. Distribution
This document is authorized for unrestricted use and release and is available in the Resources Section of the vZLC Website.

1-1-4. Cancellation
This document cancels HLN SOP, effective January 23, 2025.

Section 2. Operational Positions

Table 1-2-1. HLN ATCT Operational Positions Table

Sector/Position	STARS ID	VATSIM Callsign	Frequency
Ground Control (GC)	**N/A**	**HLN_GND**	**121.900**
Local Control (LC)	**N/A**	**HLN_TWR**	**118.300**
Approach Control (AP)	**N/A**	**HLN_APP**	**119.500**

Bold designates a primary position.

Section 3. Intrafacility Coordination Procedures

1-3-1. Runway Configurations

The active runway configuration refers to the general cardinal direction of departures and arrivals currently in use. Authorized runway configurations are listed in subparagraphs c. and d.

a. When selecting a runway configuration, the following factors must be considered:
   (1) Wind direction.
   (2) Wind speed. The gust factor takes precedence over the constant wind speed.
b. West Flow is designated as the “calm wind” runway configuration.
c. West Flow: Landing and departing Runway 27.
d. East Flow: Landing and departing Runway 9. Runway 5 may be issued at the pilot's request.
e. Runway 17/35 is a VFR-only runway.
f. Runway 10/28 is a turf seasonal runway only to be used by VFR aircraft by request only.
g. Rwy 5/23 and Rwy 17/35 are not available to commercial airliners with greater than 30 seats.

1-3-2. Changing Runway Configurations

The Local Control (LC) position is the final authority for selecting the runway configuration, but coordination with GC and Approach must be completed before changing the configuration.

a. LC must:
   (1) Coordinate with GC to determine which aircraft will be the last to depart and arrive in the old runway configuration.
   (2) Ensure that:
      (a) The ATIS is updated to reflect the runway configuration change.
   (3) Notify Approach Control (or other appropriate overlying sector) of the runway configuration change.

1-3-3. ATIS

Regardless of the position broadcasting the ATIS, the following statements must be included on the ATIS broadcast:

a. During VFR weather conditions: “DEPARTING VFR AIRCRAFT ADVISE GROUND CONTROL OF REQUESTED HEADING OR DESTINATION.”
b. When the outside air temperature is 27°C or greater: “CHECK DENSITY ALTITUDE.”

NOTE-
This statement must be made immediately after stating the altimeter setting.

c. Record the following message at the end of the ATIS: “ADVISE ON INITIAL CONTACT YOU HAVE INFORMATION (current ATIS code) WITH YOUR POSITION AND ALTITUDE.”

Section 4. Opposite Direction Operations (ODO)

1-4-1. General
Opposite Direction Operations consist of IFR/VFR operations conducted on the same or parallel runway, where an aircraft is operating in a direction opposite to another aircraft arriving, departing, or performing an approach.

NOTE-
Departing aircraft include aircraft conducting a touch-and-go, stop-and-go, low approach, missed approach, or go-around.

1-4-2. Responsibilities

a. LC and Approach Control to share the responsibility to coordinate ODO and issue traffic advisories (see below).
b. LC is responsible for applying the cutoff point between arriving and departing aircraft in opposite directions.
c. Approach Control is responsible for applying the cutoff point between aircraft arriving in opposite directions.

1-4-3. IFR Aircraft Procedures

a. General:
   (1) Traffic advisories must be issued to all ODO aircraft.

   PHRASEOLOGY-
   OPPOSITE DIRECTION TRAFFIC (distance) MILE FINAL, (type aircraft).
   OPPOSITE DIRECTION TRAFFIC DEPARTING RUNWAY (number), (type aircraft).
   OPPOSITE DIRECTION TRAFFIC (position), (type aircraft).

   (2) VFR aircraft conducting practice approaches must receive IFR services.
   (3) Opposite direction, same or parallel runway, operations with opposing traffic inside the cutoff point are prohibited.
   (4) Lateral separation must exist before any other type of separation is applied.
b. Coordination:
   (1) LC must verbally request opposite direction departures with APP.
   (2) APP must verbally request opposite direction arrivals with LC.
   (3) Initial coordination must include callsign, type, and arrival/departure runway. The phrase “opposite direction” must be used.
c. Cutoff procedures for aircraft conducting ODO to the same runway:

   NOTE-
   When both aircraft are receiving IFR services, visual separation is not authorized.
   (1) A departing aircraft must be airborne and have turned to avoid conflict prior to an aircraft reaching:

(a) A pilot reports five flying miles from the threshold of the runway of intended landing or, if an aircraft is established in the traffic pattern, prior to that aircraft turning base leg.
(b) If the above conditions are not met, action must be taken to ensure control instructions are issued to protect the integrity of the cutoff points.

(2) An arriving aircraft must cross the runway threshold prior to an aircraft reaching:
(a) A point five flying miles from the threshold of the runway of intended landing or, if an aircraft is established in the traffic pattern, prior to that aircraft turning base leg.
(b) If the above conditions are not met, action must be taken to ensure control instructions are issued to protect the integrity of the cutoff points.

d. Cutoff procedures for aircraft conducting ODO to parallel runways:
(1) Provide a turn away from the opposing arriving traffic before the arrival is inside of the cutoff point to the other runway. The same cutoff points in subparagraph c. must apply.

NOTE-
1. *Visual separation of any type may be applied after the turn away from opposing traffic is issued.*
2. *Prior to a pilot providing visual separation, a heading with at least 15° angular difference between the aircraft's flight paths must be assigned to ensure flight paths do not cross.*

1-4-4. VFR Aircraft Procedures

a. Ensure VFR aircraft are issued a turn to avoid conflict with the opposing IFR/VFR traffic.
b. During coordination, the phrase "opposite direction" must be used.
c. LC must issue traffic information to both aircraft and indicate the direction that the departure will turn (arrival/departure ODO) or the location of the opposing aircraft on final (arrival/arrival ODO).

Chapter 2. Operational Positions

Section 1. Ground Control (GC)

2-1-1. Position Responsibilities

a. Monitor the GC frequency (121.900 MHz).
b. Prepare, update, and mark flight progress strips in accordance with the ZLC Flight Strip Handling Directive (7210.3).
c. Issue departure clearances to IFR aircraft, complying with posted traffic management initiatives (TMIs), letters of agreement (LOAs), and published preferred routes.
d. Handle VFR departure aircraft in accordance with paragraph 2-1-4.
e. Ensure all taxiing aircraft have the current ATIS.
f. Determine/finalize aircraft departure runway assignments.
g. Assume other ground control or other cab responsibilities, as assigned.

2-1-2. Area of Jurisdiction

GC has jurisdiction of all movement areas except active runways. If LC releases control of Runway 5/23 and 17/35 to GC (see paragraph 2-1-6), GC assumes control of that runway for taxi use and crossings.

2-1-3. IFR Departure Instructions

Departing IFR aircraft must be assigned the following:

a. In Both Flows (West & East)
   i. All aircraft departing to the North and East are to be issued the STAKK SID.
   ii. All aircraft departing to the Northwest are to be issued the SIEBE SID.
   iii. All aircraft departing to the South are to be issued the HELENA SID.
   iv. All IFR Aircraft departing on Rwy 5 are to be issued the DIVIDE ODP.

NOTE-
If the aircraft filed a DP that is listed above, you (The Controller) Are Not Required To Change Them To The Appropriate SID based on direction but are required to change if they use RWY 5.

NOTE-
Departure headings are not authorized, as departure control is a non-radar facility.

b. An initial altitude of 9,000 feet, or lower cruising altitude if compliant with the MOCA.

2-1-4. VFR FF Departure Instructions

All VFR aircraft that requested flight following and will depart the Class D surface area must:

a. Be issued both of the following:
   (1) Appropriate departure frequency.
   (2) Discrete beacon code.
b. Have the following filled out:
   (1) Flight Strip, in accordance with the ZLC Flight Strip Handling Directive (7210.3).

(2) Flight Plan.

NOTE-
Aircraft that depart the Class D surface area and are not requesting flight following are not to have any of the above, except a filled out Half Strip.

2-1-5. Traffic Flow
GC must ensure all aircraft yield to aircraft exiting a runway, unless otherwise coordinated with LC.

2-1-6. Control of Runway 5/23 & 17/35
Runway 5/23 & 17/35 must be under the jurisdiction of LC by default. However, it is common for LC to release control of Runway 5/23 & 17/35 to GC when operationally advantageous. When GC requests control of Runway 5/23 & 17/35, verbal coordination must be accomplished in accordance with the following.

EXAMPLE-
To request control: "REQUEST CONTROL OF RUNWAY (number)."
To release control: "RUNWAY (number) RELEASED TO (position)."

2-1-7. ATIS
Ensure all departure aircraft have the current ATIS.

2-1-8. Pushback Clearances
To the maximum extent possible, aircraft parked at the terminal ramp should push back within the non-movement area.

2-1-9. Use Of SAID System

a. SAID only displays ADS-B-equipped and transmitting aircraft/vehicles within the site-specific adapted area (the airport itself, not the surrounding area).
b. SAID systems may rely solely on ADS-B transponder information to display aircraft identification if NAS flight plan information is not available to the system. This may result in a discrepancy between SAID depicted aircraft identification and NAS flight plan identification. This means that aircraft must have their ADS-B on with Altitude Reporting Enabled in order to move on the airport movement areas.
c. SAID-derived information may only be used for reference and requires confirmation via visual observation and/or pilot/operator reporting

Section 2. Local Control (LC)

2-2-1. Position Responsibilities

a. Monitor the LC frequency (118.300 MHz).
b. Provide control instructions to aircraft within the LC area of jurisdiction.
c. Determine the appropriate runway configuration in accordance with paragraph 1-3-1.
d. Sequence and separate departures and arrivals.
e. Mark FPSs in accordance with the ZLC Flight Strip Handling Directive (7210.3).
f. Assume other local control or other cab responsibilities, as assigned.

2-2-2. Area of Jurisdiction

LC has jurisdiction of the following:

a. Controls active runway(s), and controls class Delta airspace (a 4.5 mile circle from the center of the airport with 1.5 mile extensions to the east for runway 27 and to the west for runway 9, at and below 6,400 feet MSL).

2-2-3. Departures

a. Departure Notification / Releases
   i. A DM notification needs to be sent to Salt Lake Center so that Center can activate the flight plan and start radar track. That can be done via the following:
      1. `.hlndmait <Sector ID> <asel>`
   ii. Tower shall get releases from HLN Approach as well via either voice coordination or .rr command in CRC.
b. VFR aircraft to unicom, unless the VFR has requested Flight Following and those go to the appropriate departure frequency.

2-2-4. Approaches/Arrivals

a. Missed Approach/Go Around
   i. Send a Missed Notification (.MA) for the aircraft executing an unplanned Missed Approach/Go Around to the appropriate Approach sector that will be receiving the aircraft for re-sequencing.
   ii. The following is considered a Primary Climb-Out procedure and does not require any special coordination with Center or Strip marking other than the basic Missed Notification. Aircraft must not turn prior to the departure end of the runway. Handoff/Comm-Change to the appropriate Center sector must be completed no later than 1 mile off the departure end of the runway.
b. IFR: Fly the missed approach as published
c. VFR: Join the local pattern

2-2-5. Line Up And Wait (LUAW)

a. LUAW operations are not conducted at Helena ATCT.

2-2-6. Land And Hold Short Operations (LAHSO)

a. LAHSO operations are not conducted at Helena ATCT.

2-2-7. Multiple Runway Crossings For Taxiing Aircraft

a. Helena ATCT is not authorized to conduct multiple runway crossing for taxiing aircraft.

2-2-8. Turf Runway 10/28 Operations

a. Turf 10/28 is defined as a movement area.
b. Do not assign Turf 10/28 to arrivals or departures unless requested by the aircraft
c. Controllers must instruct aircraft exiting Turf 10/28 to turn left onto taxiway BRAVO or turn right onto taxiway DELTA as part of the aircraft's taxi instructions.
d. Due to the proximity of Turf 10/28 to runway 09/27, controllers must utilize intersection wake turbulence separation between aircraft departing runway 09/27 and aircraft departing the Turf 10/28 in accordance with FAA JO 7110.65 paragraph 3-9-7 when required:

NOTE-
Landing/departures on Turf 10/28 is limited to Category I single-propeller driven General Aviation aircraft and helicopters. Ensure that aircraft on final to the turf 10/28 strip do not overfly aircraft on the movement area.

e. Simultaneous, same direction operations to the Turf 10/28 and runway 9/27 are authorized in accordance with the LOA and FAA JO 7110.65.

2-2-9. Density Altitude

a. Broadcast a density altitude advisory to departing general aviation aircraft whenever the temperature is 80 F (27 C) or higher unless the advisory is included on the ATIS and the pilot reports the appropriate ATIS code.

PHRASEOLOGY-
"(Call sign) CHECK DENSITY ALTITUDE".

Chapter 3. Approach Control

Section 1. Non-Radar Standarization

3-1-1. vStrip Configuration

To ease non radar procedures, airspace within a sector is typically broken up in bays to help ease complexities of sectors. HLN has only 1 bay because the airspace is not very complex.

a. HLN Bay
   (1) Mandatory Reporting Points: None
   (2) Every Aircraft that will enter HLN approach airspace as depicted in Appendix 1, must have a flight strip in the HLN bay.
   (3) Aircraft that are proposed, must have their strip offset to show it is a proposal

Note: *vStrips does not currently automatically add strips for satellite airports. Controllers should use external applications like vatsim-radar.com to monitor for flight plans departing satellite airports.*

Section 2. Non-Radar Departures

3-2-1. Helena Airport

a. STAKK SID: Upon check in, aircraft shall be asked to report the estimated time for STAKK and joining the HLN 15 DME.
   (1) Time Approaches: Authorized with RNAV 09's and LOC-BC approaches and STAKK Departure
b. SIEBE SID: Upon check in, the aircraft shall be asked to report the estimated time for SIEBE.
   (1) Time Approaches: Authorized with RNAV 27's and ILS-Z 27 and the SIEBE Departure
c. HELENA & DIVIDE SID: Time approaches NA. If crossing traffic prevents climbs above 9,000 msl, aircraft shall be given hold over HLN VOR.
d. No SID - Not Authorized. Use DIVIDE ODP.

3-2-2. Deer Lodge Airport (38S)

a. For IFR operations at Deer Lodge Montana (38S), ZLC must coordinate with Helena ATCT at least 10 minutes prior to issuing an IFR clearance at 38S.
b. Airspace depicted in Attachment 3 must be released to ZLC as soon as practical. Upon completion of activity at 38S, ZLC must release Deer Lodge Shelf back to Helena ATCT.

Section 3. Non-Radar Arrivals

3-3-1. ZLC AIT

ZLC shall send an AIT Time for the estimated entrance into the Procedural Control Environment. This time must be entered into box <number> with the center estimated procedural control entry time. The entry fix should be displayed in box G.

3-3-2. Helena Airport

a. West Flow (Landing 27)

RNAV's are the preferred approaches for HLN, with the Z's having multiple feeder fixes on the border with center or within the TRACON.

(1) RNAV: RNAV Z or Y shall be the preferred approach. HLN APP May coordinate HIA and URELE provided ZLC provides non radar separation upon entering the TRACON. Otherwise aircraft filed over one of the other transitions shall be cleared for the approach at an appropriate altitude upon check-in. All aircraft will be asked to report Estimated time over GOKME and report passing GOKME.

(2) RNAV X: Once cleared for the approach, aircraft shall be asked to report estimated time over GASBE and report passing GASBE.

(3) Non RNAVs: If timed approaches are in use with departures utilizing the SIEBE departure, ILS Z shall be the preferred approach, otherwise there shall be no preference. Timed approaches on the ILS Z shall use 7110.65 6-7-5b 1 Minima

`7110.65 Reference 6-7-5b`

> **b.** Use the following time or radar interval as the minimum interval:
>
> **1.** Behind super:
>
> **(a)** Heavy - *3 minutes* or *6 miles.*
>
> **(b)** Large - *3 minutes* or *7 miles.*
>
> **(c)** Small - *4 minutes* or *8 miles.*
>
> **2.** Small behind heavy - *3 minutes* or *6 miles.*

b. East Flow (Landing 09)

(1) RNAV:RNAV Z or Y shall be the preferred approach. HLN APP May coordinate GLUES provided ZLC provides non radar separation upon entering the TRACON. Otherwise aircraft filed over one of the other transitions shall be cleared for the approach at an appropriate altitude upon check-in. All aircraft will be asked to report Estimated time over LISTY and report passing LISTY.

7110.65 Reference 6-7-5b

b. Use the following time or radar interval as the minimum interval:

1. Behind super:

(a) Heavy - *3 minutes* or *6 miles.*

(b) Large - *3 minutes* or *7 miles.*

(c) Small - *4 minutes* or *8 miles.*

2. Small behind heavy - *3 minutes* or *6 miles.*

3-3-3. 7S6 White Sulphur Springs

RNAV GPS 19 and 01 Are Not Authorized due to non radar separation procedures within HLN APP.

Section 4. Overflights

3-4-1. HLN

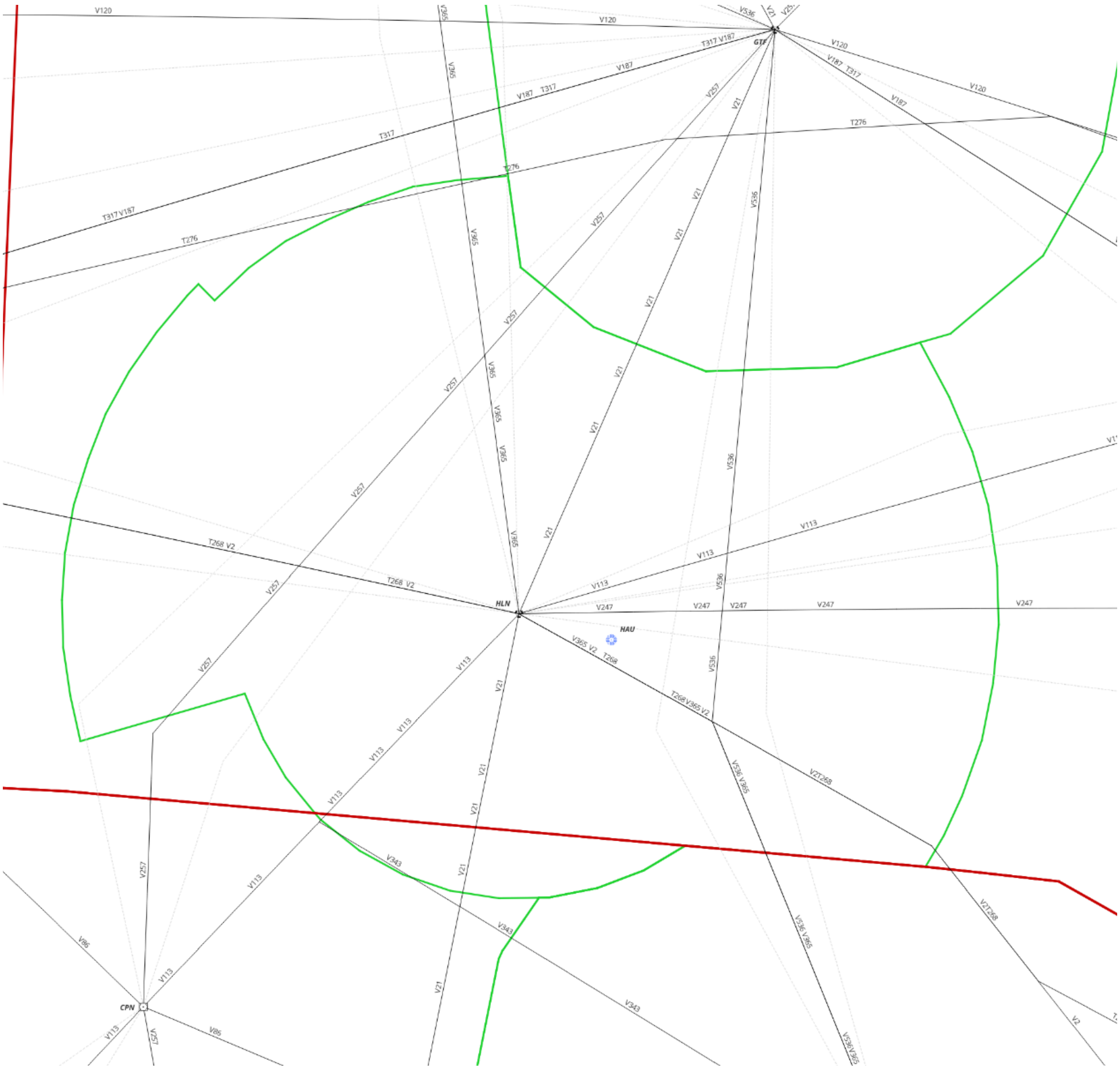
Aircraft shall be asked for estimated time over the HLN VOR upon check in. Once over HLN, the next reporting point shall be the final fix before enter ZLC airspace, which is listed below:

Airway	Fix
V247	HEXOL
V113(W of HLN)	LINGE
V21(N of HLN)	PONVE
V365(N of HLN)	WOKEN
V2(W of HLN)	PIXXI
V113(S of HLN)	EVVER
V21(S of HLN)	NNORA
V365/V2(E of HLN)	CONNS

3-4-2. Airways

a. V257
   Shall be asked for estimates and reporting times for WOKEN, RICHD, and the exit fixes SCAAT and TUCOK.
b. V536
   Shall be asked for estimates and reporting times for SWEDD, FALDE, and ANNEM, and the exit fixes GODFE and URELE
c. Aircraft southbound on V113 and V21 shall be handed off to Salt Lake Center Sector 06 on 132.400

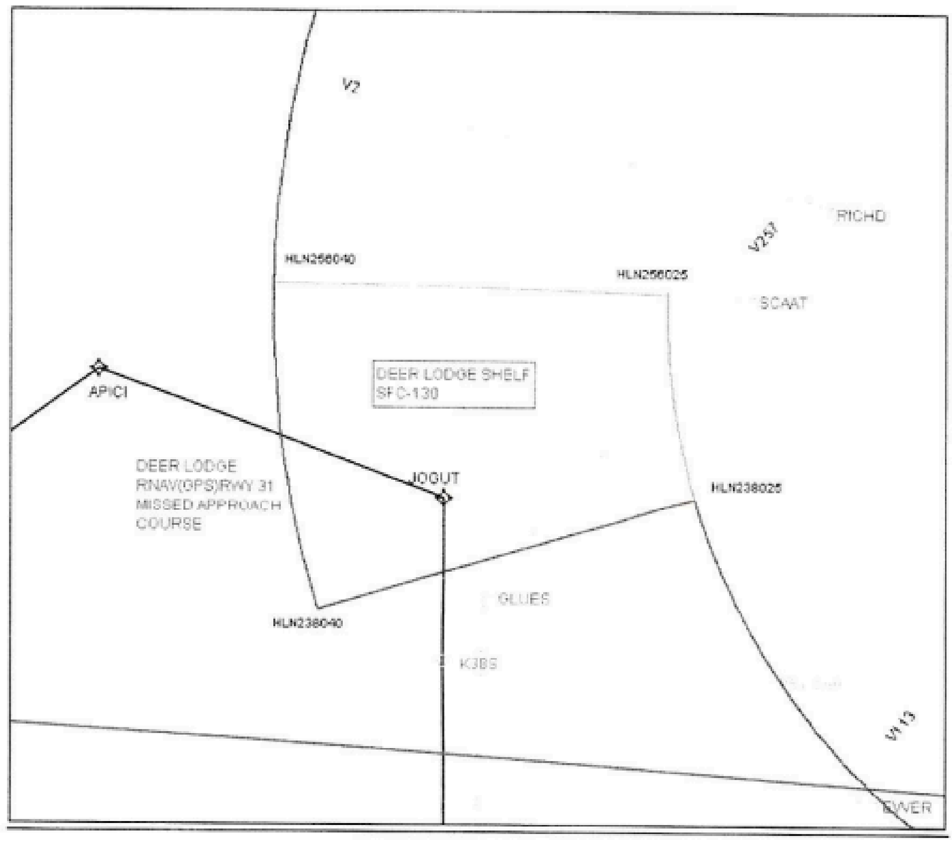
Appendix 1. Airspace



Appendix 2. Deer Lodge Shelf

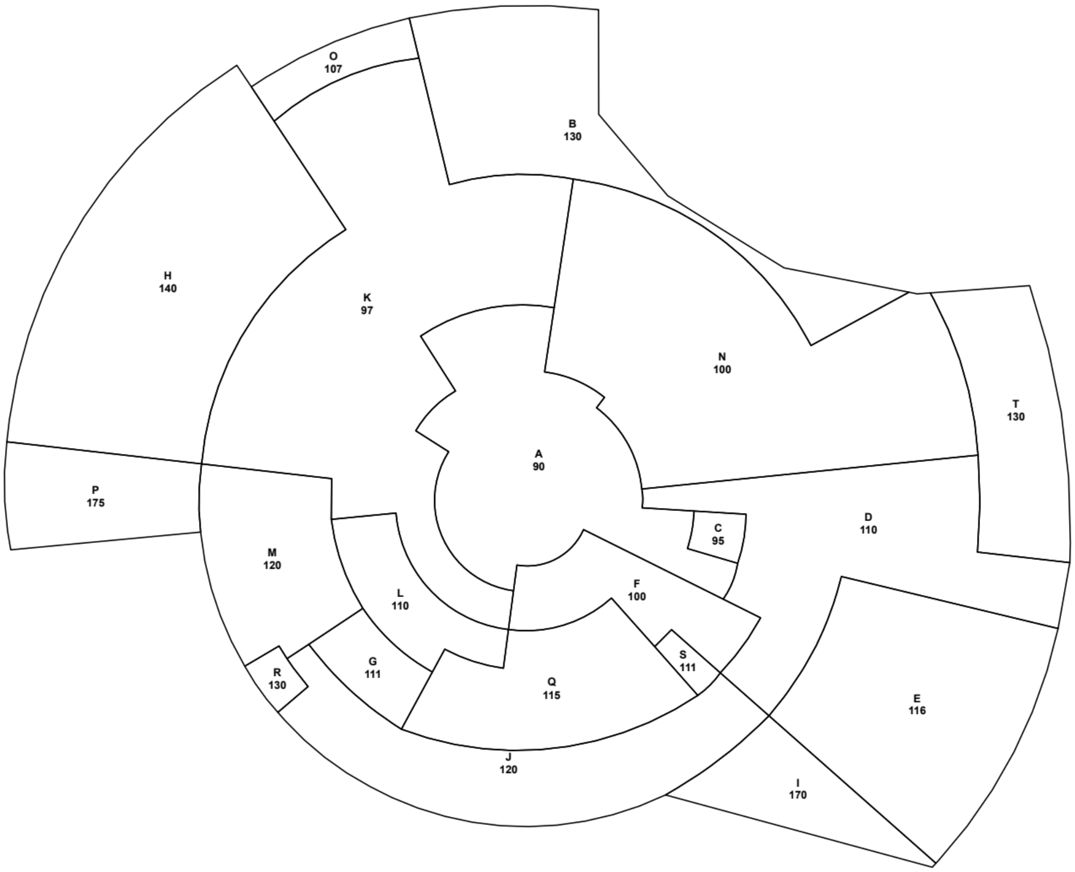
Attachment 3

<u>LATERAL BOUNDRY OF DEER LODGE SHELF</u>



FROM:
46°37′33.98″N /112°55′13″W (HLN256@40DME) TO 46°37′10.40″N /112°33′27.87″W (HLN256@25DME) TO 46°29′27.37″N /112°32′1.03″W (HLN238@25DME) VIA A 25DME ARC TO 46°25′10.75″N /112°52′47.89″W (HLN238@40DME) TO POINT OF BEGINNING VIA A 40DME ARC.

Appendix 3. Minimum Vectoring Altitude



Appendix 4. Strip Marking

Due to the non radar component for working HLN APP, the following applies IN ADDITION to ZLC Order 7210.3A

IFR Arrivals

Box 4: First two letters of the FAF

Box 5: FAF estimate time.

IFR Departures

Box 9: Estimated time joining the HLN 10 DME (SNAKO SID) or airway for non SID departures.

Box 8: HLN Release time.

IFR Overflights

Boxes 1, 4, 7: First two letters of any reporting point

Boxes 2, 5, 8: Estimated time over reporting point in adjacent boxes.

Boxes 3, 6, 9: Actual time over reporting point.